

KAESONG, Korea, Democratic People's Republic of

WMO: 470700

Lat: 37.967N

Lon: 126.567E

Elev: 230

StdP: 14.57

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	12.3	-9.1	3.4	17.3	-4.2	4.4	18.0	22.6	15.8	20.2	21.1	3.9	320	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	89.0	78.4	86.3	76.5	84.0	74.9	80.3	86.5	79.0	83.9	77.7	81.8	5.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	150.6	83.3	77.5	144.3	82.0	76.5	139.2	80.8	44.3	86.9	42.7	83.9	41.5	82.0	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	17.9	15.0	3.7	92.9	4.5	2.7	0.5	94.9	-2.1	96.4	-4.6	98.0	-7.9	99.9		
			2.7	82.9	4.3	3.0	-0.4	85.0	-2.9	86.7	-5.3	88.4	-8.4	90.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	25.9	30.4	40.5	52.1	62.6	71.2	76.3	77.6	69.4	57.8	43.3	30.1
	DBStd	18.96	7.08	6.89	6.25	6.33	4.81	3.90	3.85	4.13	4.57	6.22	7.93	7.58
	HDD50	2504	749	549	300	49	0	0	0	0	14	225	618	
	HDD65	5420	1214	969	758	388	106	4	0	15	233	650	1083	
	CDD50	3677	0	0	7	112	390	636	816	856	581	254	25	0
	CDD65	1119	0	0	0	2	30	190	351	392	146	8	0	0
	CDH74	7455	0	0	0	17	214	1002	2327	3185	680	30	0	0
	CDH80	1828	0	0	0	1	25	201	582	937	81	1	0	0
Wind	WSAvg	4.5	4.1	4.7	5.6	5.2	4.7	4.4	4.7	4.6	4.0	3.7	4.1	3.9
Precipitation	PrecAvg	53.1	0.6	1.1	1.4	2.7	4.3	5.4	16.1	11.4	5.4	2.0	1.8	0.8
	PrecMax	90.2	2.2	2.8	4.9	6.6	10.7	21.1	39.7	32.9	21.9	8.1	4.4	2.6
	PrecMin	23.6	0.0	0.0	0.1	0.4	0.4	1.1	5.5	1.3	0.2	0.1	0.4	0.1
	PrecStd	14.5	0.5	0.8	1.1	1.4	2.4	4.3	7.8	7.8	5.1	2.0	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	53.4	64.3	76.4	83.2	87.7	92.7	92.5	84.8	77.7	65.7	50.3
		MCWB	40.2	45.5	51.5	58.3	65.6	71.5	79.3	80.4	71.6	64.5	57.8	44.3
	2%	DB	41.6	47.3	58.1	70.7	79.1	84.5	88.2	90.0	82.5	73.6	61.2	46.9
		MCWB	36.7	40.0	48.3	57.1	64.0	70.4	78.0	79.5	71.2	62.2	54.3	42.5
	5%	DB	38.7	43.7	54.3	66.9	75.9	82.2	85.4	87.5	80.3	70.8	58.0	44.0
		MCWB	34.4	37.8	45.7	54.8	63.0	69.8	76.8	78.3	70.3	60.6	52.5	39.5
	10%	DB	36.4	40.6	50.9	63.5	72.9	79.8	83.0	85.0	78.2	68.0	55.2	41.2
		MCWB	32.6	35.6	43.3	53.1	61.8	68.9	75.5	77.1	68.7	58.7	50.3	37.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	47.9	54.2	63.6	69.8	75.1	82.1	82.7	77.1	67.4	60.1	46.7
		MCDB	44.4	51.3	60.2	70.1	77.6	82.5	88.6	88.5	81.0	73.7	63.5	49.0
	2%	WB	37.9	41.6	50.2	59.5	66.8	73.4	79.9	80.9	74.7	64.6	56.5	43.0
		MCDB	40.4	45.9	56.2	67.4	74.7	80.7	85.8	87.2	79.0	70.0	59.5	46.2
	5%	WB	35.3	38.5	47.0	57.0	65.1	72.0	78.3	79.8	72.8	62.8	53.8	40.2
		MCDB	37.9	42.7	53.0	64.8	72.5	78.3	83.1	85.3	77.2	68.4	57.0	43.3
	10%	WB	33.0	36.1	44.3	54.6	63.4	70.8	77.1	78.6	71.1	60.9	50.9	37.6
		MCDB	35.9	40.1	49.7	61.6	70.0	76.4	81.0	83.2	75.4	66.1	54.4	40.8
Mean Daily Temperature Range	5% DB	MDBR	15.0	15.8	16.2	17.9	17.6	14.8	10.5	12.0	15.4	17.8	15.2	13.9
		MCDBR	16.4	18.6	21.0	22.4	22.8	19.4	14.7	15.4	17.8	20.1	18.0	15.6
		MCWBR	12.6	12.5	12.7	11.3	10.8	8.4	7.5	7.0	8.4	10.3	12.1	12.2
	5% WB	MCDBR	14.4	17.0	19.0	19.2	18.5	15.6	12.1	12.9	14.1	17.1	15.3	14.8
		MCWBR	12.6	12.9	12.7	11.5	10.5	7.9	7.1	6.9	8.5	10.2	12.3	12.6
Clear-Sky Solar Irradiance	taub	0.420	0.480	0.557	0.610	0.621	0.695	0.667	0.573	0.504	0.501	0.478	0.423	
	taud	1.939	1.840	1.719	1.655	1.681	1.598	1.707	1.923	2.027	1.911	1.908	1.963	
	Ebn at Noon	224	227	223	221	222	206	210	228	236	218	203	212	
	Edh at Noon	44	55	69	77	77	83	75	59	50	51	45	40	
All-Sky Solar Radiation	RadAvg	797	1032	1356	1577	1773	1659	1291	1390	1335	1135	765	689	
	RadStd	48	78	125	110	174	179	193	190	137	75	75	48	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	+103

Nomenclature: See separate page